Linux Fundamentals – Lesson 2: Linux Architecture

Type	Draft
Status	Not started
Project	My DevOps Journey 🚀

First, what does architecture mean?

Architecture means how something is built and how its different parts work together.

Linux also has different parts that work together.

1. Main parts of Linux Architecture

Linux has 4 important parts:

1. Hardware

Hardware means the physical parts of a computer.

Examples:

- CPU
- RAM
- Hard disk
- Keyboard
- Mouse

Simple meaning: Hardware is the part of the computer that we can touch.

2. Kernel

The kernel is the main part of Linux.

It connects software with hardware.

Its work:

- Manages memory (RAM).
- Manages the CPU.
- Controls hardware.
- Manages running programs.

Simple meaning: Kernel manages the computer's resources.

3. Shell

Shell helps us communicate with Linux using commands.

For example:

```
pwd
```

When we type this command, the shell understands it and asks Linux to perform the work.

Simple meaning: Shell takes our commands and communicates with the operating system.

4. Applications and Utilities

These are the programs and tools we use on Linux.

Examples:

- Text editors
- File managers
- Web browsers
- Linux commands such as `ls` and `mkdir`

Simple meaning: These are the tools and programs that help us do our work.

2. How do these parts work together?

Let's take a simple example.

You type this command:

```
pwd
```

What happens?

1. You type the command in the terminal.
2. Shell understands the command.
3. Linux performs the required work.
4. The result is displayed on your screen.

This is how different parts work together.

3. Important points for your notes

Part	Work
Hardware	Physical parts of computer
Kernel	Manages hardware and system resources
Shell	Takes commands from users
Applications	Programs and tools used by users

Remember this order:

Hardware → Kernel → Shell → User commands and applications.